

Non-additive Strain–Dopant Energetics in Quasi-Hexagonal C₆₀ Graphullerene

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Under mechanically constrained loading on quasi-hexagonal (qHP) C₆₀ graphullerene, measurable deviations from additive strain–dopant energetics arise because frozen elastic incompatibility introduces a bilinear strain–composition interaction that can reach magnitudes comparable to high-throughput screening margins—not because a finite-difference estimator is merely nonzero. The additive approximation in sequential workflows corresponds to neglecting the mixed second derivative $\partial^2 E/\partial\epsilon\partial\delta$ in the total-energy functional; matched four-corner audits on B/N/P tetramers and periodic $n \times \text{C}_{60}$ supercells ($n = 1\text{--}8$, PBE+D3 rigid protocol) show that this term is large enough to change stability rankings for size-mismatched phosphorus ($|\mathcal{S}| = 31.9 \pm 2$ meV/atom at $n=1$, $\eta \approx 8\%$ at $n=4$). The four-corner difference $\mathcal{S}$ estimates that coupling [Eqs. (4)–(3)] and is not a new thermodynamic observable. Non-additivity scales with frozen local elastic incompatibility ($|\Delta r_{\text{cov}}| \gtrsim 20$ pm) rather than dopant identity alone; ionic relaxation nearly extinguishes $|\mathcal{S}|$, consistent with a dominant elastic-work contribution under constrained load rather than a partitioned claim of intrinsic electronic cross-coupling. The audit signals when sequential additive workflows should be replaced by explicit coupled evaluations on the $E(\epsilon, \delta)$ landscape.

I. INTRODUCTION

Substitutional doping and mechanical strain are widely used strategies for tuning energetic stability and electronic structure in covalent molecular networks [1, 2]. The additive approximation used in sequential high-throughput screening corresponds to neglecting the mixed second derivative $\partial^2 E/\partial\epsilon\partial\delta$ in the total-energy functional; the physical question is when that omitted bilinear term reaches magnitudes comparable to meV-per-atom stability margins under realistic loading boundary conditions. While external biaxial strain modifies global bonding geometry, substitutional dopants introduce localized chemical stress, lattice distortion, and bond competition, so the two perturbations can become energetically coupled through higher-order cross terms in the total-energy expansion [3]—i.e., the total response need not separate into strain-only and composition-only increments without a nonlinear interaction term. Configurational interaction and cluster-expansion effective Hamiltonians for alloys already include explicit composition–strain cross terms [3], so a bilinear correction to $E(\epsilon, \delta)$ is expected in principle. Expanding the energy per atom $e(\epsilon, x) = E(\epsilon, \delta)/N_{\text{atoms}}$ about the pristine reference gives, schematically,

$$e(\epsilon, x) \approx e_0 + a\epsilon + bx + c\epsilon x + \mathcal{O}(\epsilon^2, x^2, \epsilon^2 x, \epsilon x^2), \quad (1)$$

where $c\epsilon x$ is the lowest-order non-additive coupling between external strain and composition; at finite load, $c = \partial^2 e/\partial\epsilon\partial x$ is the bilinear coefficient whose relevance we test. Although dopant composition is sampled discretely (one substituent per C₆₀ on a fixed map), Eq. (1) is interpreted as a local finite-difference representation of the underlying composition-dependent energy functional, not as a strict differential expansion in a continuous concentration field. The continuum coef-

ficient $\gamma \equiv \partial^2 E/\partial\epsilon\partial\delta$ characterizes the intrinsic mixed response; the four-corner estimator $\mathcal{S}$ (Sec. II C) samples that response at +3% biaxial load—a diagnostic, not the physical claim (full derivation: Supplemental Material, Sec. I). In many high-throughput mapping studies, strain and substitution are nevertheless evaluated in separate density-functional theory (DFT) workflows and combined additively, in part because exhaustive coupled (ϵ, δ) grids are computationally expensive [1, 4, 5]. Although first-principles work has extensively mapped strain and doping separately across molecular and two-dimensional networks [5–7], matched four-corner quantification of their total-energy coupling under simultaneous load has rarely been reported for fullerene-based covalent molecular networks. We study quasi-hexagonal (qHP) C₆₀ graphullerene [8, 9] as an experimentally accessible model covalent molecular network. Prior density-functional work largely falls into two classes: additive mapping studies that sample strain and substitution on separate grids [5–7, 9, 10], and reports of nonlinear strain–composition response in band edges, transport, or magnetism without matched four-corner *total-energy* audits [4, 11, 12]. A literature-positioning summary is given in Supplemental Material, Table I. What remains unresolved for soft graphullerene cages is how large the bilinear correction is, when it reverses *relative energetic stability* and dopant-site preference under simultaneous load—quantities that govern processing and defect retention rather than gap or mobility shifts alone—and when sequential additive workflows become unreliable. Ab initio models of pristine qHP networks report intrinsic intramolecular strain [13]. A substitutional defect imposes a local chemical-stress and bond-distortion field on a prestrained, elastically soft cage; together with global biaxial load, this motivates a periodic-scale audit of dopant–strain coupling beyond linear α .

Here we demonstrate that additive strain–dopant energetics fail under mechanically constrained joint load on qHP C_{60} because frozen elastic incompatibility generates a bilinear strain–composition interaction comparable to screening energy scales—not merely because a four-corner difference is nonzero. On the total-energy landscape $E(\epsilon, \delta)$ under biaxial load—where *relative energetic stability*, not band-edge shifts alone, governs accessible substitutional configurations—we test whether (i) additive totals fail under joint (ϵ, δ) load; (ii) the omitted correction matters under mechanically constrained configurations; and (iii) failure tracks covalent-radius mismatch rather than donor/acceptor π -electron descriptors alone. We focus on qHP C_{60} as an experimentally accessible, elastically soft model covalent network [8, 13]; substitutional B, N, and P provide a minimal donor, acceptor, and size-mismatch triad (~ 7 – 30 pm covalent-radius excess vs. C; Table VII). Matched four-corner audits quantify the omitted bilinear term; $\mathcal{S}$ [Eq. (2)] reports its meV-per-atom value and is *not* proposed as a new physical quantity (Sec. II C). The primary discovery is *physical*: stored elastic incompatibility under constrained load can break additive high-throughput workflows—not a new descriptor or thermodynamic order parameter (Sec. IV B). On qHP C_{60} , the audit protocol is transferable only after repeating the same (ϵ, δ, n) grid on each host; we report failure conditions on one network, not materials-general laws.

II. COMPUTATIONAL METHODS

A. Electronic structure method

All total-energy calculations use CP2K [14] with Goedecker–Teter–Hutter (GTH) pseudopotentials, the DZVP-MOLOPT-SR-GTH atom-centered basis, and the Gaussian and plane waves (GPW) scheme.

a. Exchange-correlation and dispersion. We use PBE [15] with Grimme DFT-D3 (Becke–Johnson damping) [16] for the production dataset. Hybrids or meta-GGAs (SCAN, r^2 SCAN) can shift absolute totals by tens of meV per atom in carbon networks; production rankings are therefore framed as *within-PBE+D3* comparisons of $\mathcal{S}$ among B, N, and P under matched load. The four-corner construction [Sec. II C] subtracts totals computed with identical XC, basis, dispersion, and numerical settings, so systematic shifts largely cancel when *ranking* dopants at fixed protocol—but cancellation is not assumed without audit: on the tetramer P grid, Grimme-D3 non-additivity contributes only $\mathcal{S}_{\text{vdW}} \approx +0.33$ meV/atom (Table I), far below $|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}|$ (Table V). Representative SCAN and r^2 SCAN four-corner spot-checks on periodic P at $n=1$, +3% are targeted sensitivity tests (Table I; Limitations); no meta-GGA $\mathcal{S}$ values appear in the main text without converged outputs.

b. Plane-wave mesh and Brillouin-zone sampling. Plane-wave cutoffs are 400 Ry for $n \leq 4$ and 350 Ry

for $n \geq 6$ (REL_CUTOFF 50 Ry; QS EPS_DEFAULT $10^{-10} E_h$). Comparisons at fixed n always use the same cutoff. Production supercells contain 60–480 atoms with in-plane lattice constants up to ~ 44 Å at $n=8$; inputs use Γ -only sampling. Because $\mathcal{S}$ is a second-order difference on the *same* supercell, uniform k -point or cutoff shifts that co-drifts all four corners largely cancel in the four-corner estimator; sensitivity must be audited on $\mathcal{S}(\text{cutoff})$ and $\mathcal{S}(k)$, not on bare totals alone. A partial cutoff audit at $6 \times C_{60}$ N and +3% shifts one corner by only ~ 0.057 meV/atom between 350 and 400 Ry, implying $|\Delta \mathcal{S}| < 0.2$ meV/atom when only the mesh drifts (Table I). Dense- k ($2 \times 2 \times 1$) four-corner benchmarks on $n=1$ B/N/P at +3% are pending (Table I); until they converge, the smallest reported $|\mathcal{S}|$ values (2.3–3.0 meV/atom for N and B at $n=1$) are interpreted with explicit Γ -only caution.

c. SCF convergence and four-corner precision. Unless noted otherwise, the SCF threshold is EPS_SCF = $10^{-6} E_h$ in inner and outer orbital-transformation loops (MAX_SCF 300; OUTER_SCF MAX 30). Charged-state workflows in the Supplemental Material tighten this to $10^{-7} E_h$; selected stiff tetramer corners may use $10^{-5} E_h$ when outer SCF cycling persists at 10^{-6} (Table V). Because $\mathcal{S}$ is built from four independently converged totals [Eq. (2)], uncertainty is propagated as $\sigma_{\mathcal{S}} = 2\sigma_e$ for uncorrelated per-atom corner errors σ_e [Eq. (5); Supplemental Material, Sec. I E] (Table I). We use $\sigma_{\mathcal{S}} \lesssim 2$ meV/atom as the conservative reporting band on the $n \leq 4$ production grid.

d. Spin, charge, and mechanical boundary conditions. Substitutional periodic cells are charge-neutral closed-shell Kohn–Sham calculations (no spin polarization in the production $\mathcal{S}$ grid); Hirshfeld strain-path audits on $n=1$ B/N/P use spin-polarized KS only where odd electron counts require it (Sec. II D 3). We do not apply compensating background charge or charged supercells in the $\mathcal{S}$ audit: doping is modeled as neutral substitution, and ΔE_{sub} is not a formation enthalpy (Sec. II B).

Biaxial strain scales lattice vectors at fixed fractional coordinates (rigid-strain protocol). This is a deliberate mechanical *upper bound* relevant when ionic relaxation is kinetically restricted—epitaxial clamping on a rigid substrate, strong interfacial adhesion, or ultrafast biaxial loading—rather than an equilibrium free-energy prediction under fully relaxed internal coordinates (Sec. II D; Discussion, constrained-loading paragraph).

B. Strain and doping models

Biaxial strain is defined as $\epsilon = (a - a_0)/a_0 \times 100\%$, where a (a_0) is the strained (pristine equilibrium) in-plane lattice constant; dopant species $\delta \in \{\text{B}, \text{N}, \text{P}\}$. The strain–dopant coupling energy $\mathcal{S}$ is defined in Sec. II C [Eqs. (2)–(3)]; finite-difference derivation and stencil justification are in Supplemental Material, Sec. I. π -projected densities of states are written π -DOS.

Tetramer models ($4 \times \text{C}_{60}$) substitute B, N, or P at nominal ~ 5 at.% on a fixed pseudorandom carbon-site map; configurations are not thermodynamically optimized. Periodic supercells place one substituent per C_{60} (~ 1.67 at.%) on a fixed production site map shared across $n = 1-8$. Linear strain sensitivity $\alpha = dE/d\epsilon$ (meV per 1% biaxial strain) is obtained from rigid PBE+D3 tetramer grids; positive α means tensile strain increases the total energy. The substitution energy difference $\Delta E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ is not a thermodynamic formation enthalpy (no elemental chemical potentials). Reported α and $\mathcal{S}$ under the rigid-strain protocol (below) are upper bounds on equilibrium magnitudes when ionic relaxation is allowed (Sec. IID).

a. Rigid vs. relaxed mechanical protocols. *Rigid strain* scales in-plane lattice vectors at fixed fractional atomic coordinates: ionic positions do not relax with ϵ . This models mechanically clamped or ultrafast-loaded configurations and yields an upper bound on $|\mathcal{S}|$ when internal stress cannot relax. *Relaxed strain* allows fixed-cell BFGS geometry optimization at each corner before the coupling energy is evaluated on matched coordinates (Table V; Fig. 1(e)); ionic relaxation can reverse or nearly extinguish rigid $\mathcal{S}$ (Sec. IID 2). Force convergence targets are 4.5×10^{-4} and 3.0×10^{-4} Ha/bohr (~ 0.023 and ~ 0.015 eV/Å) at $\epsilon=0$ and $+3\%$, respectively, with matched displacement thresholds (Sec. IID 2); these are tighter than typical 0.02 eV/Å loose-scan defaults because relaxation can drive $|\mathcal{S}|$ toward zero and must be resolved at sub-meV/atom scale.

C. Strain–dopant coupling energy

The physical hypothesis is the bilinear cross term in Eq. (1) (Introduction): under constrained load, $c \neq 0$ and can be large enough to invalidate additive screening. The continuum coefficient $\gamma \equiv \partial^2 E / \partial \epsilon \partial \delta$ per atom (equivalently c in Eq. (1)) characterizes the *intrinsic* mixed energetic response of the host under stated boundary conditions. $\mathcal{S}$ is the symmetric four-corner *finite-difference estimator* of that response at $+3\%$ load—not a postulated order parameter and not interchangeable with γ unless higher-order remainders are negligible (descriptor comparison: Supplemental Material, Table II; full derivation: Supplemental Material, Sec. I).

a. Taylor expansion and discrete composition. Eq. (1) is a local expansion of the energy per atom $e(\epsilon, x)$ about the pristine reference. Although dopant composition is sampled discretely (one substituent label per C_{60} on a fixed map), the expansion is interpreted as a finite-difference representation of the underlying $E(\epsilon, \delta)$ functional on the audited stencil, not as a strict differential expansion in a continuous concentration field. It requires sufficiently small strain and a single smooth energy branch; at large strain ($\gtrsim 10\%$) or across phase transitions, Eq. (1) no longer applies (Sec. IV F).

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (2)$$

with $\Delta E(\epsilon, \delta) = [E(\epsilon, \delta) - E(0, \text{pristine})]/N_{\text{atoms}}$ and $\delta=0$ denoting the pristine reference at the same supercell size $n \times \text{C}_{60}$. Equivalently, on corners $(0, 0)$, $(\Delta\epsilon, 0)$, $(0, \Delta\delta)$, $(\Delta\epsilon, \Delta\delta)$,

$$\mathcal{S}(\epsilon, \delta) = \frac{E(\epsilon, \delta) - E(\epsilon, 0) - E(0, \delta) + E(0, 0)}{N_{\text{atoms}}}, \quad (3)$$

which depends only on the four corner totals (reference independence).

b. Why a four-point stencil? Among relaxed free-energy differences ΔG , enthalpic formation energies with chemical potentials, or pairwise binding increments, the symmetric four-corner form in Eqs. (2)–(3) is the minimal rectangular-grid stencil that (i) uses four DFT totals, (ii) is second-order accurate in $(\Delta\epsilon, \Delta\delta)$ for the bilinear term, (iii) cancels all first-order strain and composition increments and any shared energy offset, and (iv) equals the additive-screening bias by construction. It is not unique: central-difference, least-squares polynomial fits, or full mixed Hessians on continuous composition fields are alternative estimators of $\partial^2 E / \partial \epsilon \partial \delta$ with different cost, bias, and interpretability (Supplemental Material, Sec. IC; Table II). Asymmetric three-point schemes retain linear contamination; alternative asymmetric stencils differ from Eq. (3) by $\mathcal{O}(\Delta\epsilon, \Delta\delta)$.

c. Finite-difference link to the mixed derivative. Expanding each corner total to second order and forming Eq. (3) cancels $E_\epsilon \Delta\epsilon$ and $E_\delta \Delta\delta$ identically (Supplemental Material, Sec. IB), yielding

$$\mathcal{S}(\epsilon, \delta) \approx \frac{\partial^2 E}{\partial \epsilon \partial \delta} \frac{\Delta\epsilon \Delta\delta}{N_{\text{atoms}}} + \mathcal{O}(\Delta\epsilon^2, \Delta\delta^2, \Delta\epsilon^2\delta, \epsilon\delta^2), \quad (4)$$

so $\mathcal{S}$ reports the discrete mixed second derivative of total energy on the audited (ϵ, δ) grid. At finite $+3\%$ biaxial load and discrete B/N/P labels, higher-order terms in Eq. (4) are not separately resolved on the production periodic map; multi-amplitude strain audits ($\epsilon = 1-3\%$) are targeted sensitivity tests (Limitations). Throughout the main text, “nonlinear coupling” refers to a nonzero mixed response estimated by $\mathcal{S}$, not to $\mathcal{S}$ itself as a fundamental constant.

d. Units, symmetry, and gauge invariance. $\mathcal{S}$ is reported in meV per atom; η [Eq. (6)] is dimensionless; $\gamma = \partial^2 E / \partial \epsilon \partial \delta$ carries units of energy per atom per strain increment per composition increment, with $\mathcal{S} \approx \gamma \Delta\epsilon \Delta\delta$ on the production grid up to truncation error. The symmetric stencil is invariant under exchanging the order of the ϵ and δ increments on the grid when mixed partials commute (Clairaut symmetry; Supplemental Material, Sec. ID)—a property of the estimator, not a claim that strain and chemical substitution are interchangeable controls. $\mathcal{S}$ is invariant under arbitrary uniform energy offsets shared by all four corners. Strain ϵ and dopant label δ are distinct physical control axes in the production protocol.

e. Error propagation. If each corner carries uncorrelated equal per-atom uncertainty σ_e ,

$$\sigma_S^2 = 4\sigma_e^2, \quad \sigma_S = 2\sigma_e, \quad (5)$$

from the four-term difference in Eq. (3) (derivation: Supplemental Material, Sec. IE). Table I lists demonstrated and pending contributions to the reporting band; we use $\sigma_S \lesssim 2$ meV/atom as a conservative protocol floor on the $n \leq 4$ grid when interpreting the smallest periodic points (2.3–3.0 meV/atom for N and B at $n=1$). Periodic $\mathcal{S}$ values in the main text are from the +3% size-scaling grid (Table II).

We report $\mathcal{S}$ in meV per atom; $\mathcal{S} < 0$ (> 0) means the coupled corner is more (less) stable than the additive estimate.

To quantify additive failure magnitude we report

$$\eta \equiv \frac{|\mathcal{S}|}{|\Delta E_{\text{coupled}}|}, \quad (6)$$

where $\Delta E_{\text{coupled}} = (E(\epsilon, \delta) - E(0, 0))/N_{\text{atoms}}$. At $n=4$ and +3% strain, $\eta \approx 8\%$ for P ($|\mathcal{S}| = 23.7$ meV/atom vs. $|\Delta E_{\text{coupled}}| \approx 283$ meV/atom), whereas B and N remain below 1%.

f. Link to mechanical protocols (Secs. IID 1 and IID 2). All $\mathcal{S}$ values quoted for screening comparisons use the rigid-strain protocol unless Table V or Fig. 1(e) states otherwise. Relaxed-strain $\mathcal{S}$ is evaluated on ionically optimized coordinates at fixed biaxial cell shape; it can reverse sign relative to the rigid upper bound because internal cage stress is partially released before the four-corner difference is taken.

D. Validation benchmarks

Four protocol checks govern interpretation: (i) **four-corner numerical hygiene**—Table I (σ_S propagation, partial cutoff and vdW audits); (ii) **ionic relaxation**—tetramer P four-corner fixed-cell BFGS and periodic $n=1$ P geometry optimization (Table V; Fig. 1(e)); (iii) **matched-functional rigid vs. relaxed** coordinates on the tetramer P grid (Table V); and (iv) **alternate dopant placement** on independent site maps (Table II).

Tetramer corners use the BFGS thresholds quoted above (Sec. IID 2). Periodic ionic benchmarks include $n=1$ P on the production map (largest rigid $|\mathcal{S}|$); tetramer Table VII supplies the complementary ~ 5 at.% relaxation geometry. Periodic $n=4$ P fixed-cell relaxation, B/N periodic relaxations, and dense- k / meta-GGA four-corner spot-checks remain targeted backlogs (Table I; Limitations). Size scaling extends to $n=8$ (480 atoms, in-plane lattice ~ 44 Å). With one substituent per C_{60} and vdW-separated cages, electrostatic dilution and lateral periodicity make spurious cell–cell coupling subdominant to cage-local misfit stress on the audited maps,

TABLE I. Error budget for four-corner $\mathcal{S}$ [Eq. (3)]; SCF propagation follows Eq. (5) when corner errors are uncorrelated and equal. $\sigma_S \approx 2\sigma_e$ assumes uncorrelated per-atom corner errors σ_e ; identical XC, k -point, and cell settings across all four corners induce partial cancellation, so realized noise is often smaller. Rows marked “pending” are targeted sensitivity tests (Limitations); no meta-GGA or dense- k numbers appear without converged outputs. The reporting band (± 2 meV/atom) is a conservative protocol floor, not a quadrature sum of independent rows.

Source	$ \Delta \mathcal{S} $ or σ_S (meV/atom)
SCF convergence (EPS.SCF = 10^{-6})	$\lesssim 2$ ($\sigma_S \approx 2\sigma_e$)
Plane-wave cutoff (350 vs 400 Ry)	< 0.2
D3 dispersion cross term	$ \mathcal{S}_{\text{vdW}} \approx 0.33$
D3 dispersion (B/N periodic)	—
Ionic relaxation (rigid vs. relaxed P)	$\gg 2$ (orders of magnitude)
k -point (Γ vs $2 \times 2 \times 1$)	—
XC (PBE+D3 vs SCAN / r ² SCAN)	—
<i>Adopted reporting band ($n \leq 4$): ± 2 meV/atom.</i>	

but explicit two-body periodic-image convergence tests on $|\mathcal{S}|$ beyond this grid were not performed.

The task-group inventory in Table V (superseded exploratory grids in Table IV) lists N_{conv} and main-text linkage; ionic-relaxation energies appear in Table VII; the alternate-placement grid in Table II spans the full B/N/P strain path and six modern PBE+D3 pristine references. Three validation benchmarks—ionic-relaxation tetramer $\mathcal{S}_P$ (Table VII), alternate dopant placement (Table II), and a cutoff benchmark at $6 \times C_{60}$ N @ +3% (Table V)—govern interpretation of size-scaling trends (Secs. III A and IV).

Sec. III C reports E_{sub} only when tied to the verified PBE+D3 grids above (no chemical-potential reference; not a formation enthalpy).

1. Alternate-placement benchmark

An independent dopant configuration at the same nominal ~ 5 at.% loading repeats the B/N/P strain grid under rigid PBE+D3 (Table VI). Main-text quantitative α and $\mathcal{S}_{+3\%}^{\text{tet}}$ use the reference-placement map in Fig. 3(a,b); the alternate seed 137 grid is archived only. At periodic $n=4$, two independent site maps repeat the four-corner Eq. (2) audit; the $|\mathcal{S}|$ ranking at +3% matches the production map in Table II.

2. Ionic-relaxation benchmark

Fixed-cell BFGS geometry optimization is performed on the tetramer pristine/ P corners at $\epsilon=0$ and +3% (force thresholds 4.5×10^{-4} and 3.0×10^{-4} Ha/bohr, ~ 0.023 and ~ 0.015 eV/Å; displacement thresholds 3.0×10^{-3} and 1.5×10^{-3} bohr). These thresholds are tighter than typical 0.02 eV/Å defaults because relaxation can

TABLE II. Periodic coupling-energy audit at +3% biaxial strain (PBE+D3 rigid protocol, reference placement; core $n \leq 4$ at 400 Ry). $\mathcal{S}$ in meV/atom from Eq. (2); conservative protocol uncertainty ± 2 meV/atom for $n \leq 4$ (Sec. II C). Substitutional loading is one dopant per C_{60} unit (~ 1.67 at.% of total atoms in each supercell). Rows with $n \geq 6$ (350 Ry) are in Supplemental Material Table S1.

n (C_{60} units)	Dopant	ϵ (%)	at. %	$\mathcal{S}$ (meV/atom)
1	B	+3	1.67	-8.38
1	N	+3	1.67	+2.30
1	P	+3	1.67	-31.91
2	B	+3	1.67	-2.18
2	N	+3	1.67	+6.85
2	P	+3	1.67	-28.29
4	B	+3	1.67	-6.23
4	N	+3	1.67	+2.95
4	P	+3	1.67	-23.71

TABLE III. DFT calculation groups supporting the main-text claims (reproducibility inventory; Supplemental Material only). Superseded exploratory grids appear in Table IV.

Calculation group	System	Count	Protocol
Archived baseline (legacy PBE)	$4 \times C_{60}$	24	Rigid strain; not used for main-text claims
π -DOS and gaps	$4 \times C_{60}$	12	PBE+D3
IPR (polaron dimer)	Dimer	4	Pristine/B; $\epsilon=0, +3\%$
Hopping J	Dimer	4	PBE+D3
Charged polarons ^a	$4 \times C_{60}$	20	UKS; geometry opt. + vertical SP
Periodic size scaling	$n \times C_{60}$	41	PBE+D3; 400/350 Ry
Cutoff benchmark	$6 \times C_{60}$ N	1	+3%; 400 Ry single-point
Ionic relaxation	$4 \times C_{60}$	4	Fixed-cell GEO; P corners
Ionic relaxation	$1 \times C_{60}$ P	4	Fixed-cell GEO; periodic
Alternate placement	$4 \times C_{60}$	24	Rigid PBE+D3
Periodic placement	$4 \times C_{60}$	16	$n=4$; two independent maps
Matched-functional rigid	$4 \times C_{60}$	4	PBE+D3 at relaxed coordinates
Reference placement	$4 \times C_{60}$	24	Rigid PBE+D3
Hirshfeld population	$1 \times C_{60}$ B/N/P	18	Spin-polarized KS

^aSix relaxed tetramer geometries provide charged-state initial structures.

drive $|\mathcal{S}|$ toward zero and must be resolved at sub-meV/atom scale. A matched-functional rigid PBE+D3 control at the relaxed coordinates quantifies how $\mathcal{S}$ changes upon relaxation (Table V). The same BFGS thresholds are used for the periodic $n=1$ P four-corner relaxation reported in Fig. 1(c).

3. Hirshfeld population along strain

Hirshfeld atomic charges are computed along $\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$ on periodic $n=1$ B/N/P cells (spin-polarized KS for odd electron counts). They complement geometric $\bar{d}$ in Table VII.

III. RESULTS

We report total-energy audits on matched periodic supercells and on the reference tetramer map under mechanically constrained joint load (Sec. II C). Observations below are stated without mechanistic interpreta-

tion; Sec. IV addresses origins, screening limits, and literature context.

A. Non-additive strain-doping coupling

Figure 1 summarizes the central test: separate strain-only and substitution-only DFT increments are summed additively, compared with the explicitly coupled four-corner total, and the signed gap $\mathcal{S}$ [Eq. (3)] quantifies when that additive workflow mis-ranks stability (panel f). Panel (c) compares, at $n=4$ and +3% biaxial strain, the additive prediction $\Delta E_{\text{strain}} + \Delta E_{\text{dop}}$ with the explicitly coupled four-corner increment; their signed difference is $\mathcal{S}$. The coupled bars exceed the additive bars for B, N, and P; P shows the largest gap ($\mathcal{S} = -23.7 \pm 2$ meV/atom; $\eta \approx 8\%$). Panels (a,b) map $\mathcal{S}(n, \delta)$ on the production grid and size-scaling trajectories with provisional $S_{\infty} + A/n$ guides. Across fifteen periodic +3% points [Fig. 1(d)], $|\mathcal{S}|$ spans 2.2–31.9 meV/atom (median 7.8 meV/atom; quartiles marked); the P value at $n=1$ (31.9 ± 2 meV/atom) is an extremum within this distribu-

TABLE IV. Exploratory DFT grids superseded by the production protocol (reproducibility archive only).

Task group	System	N_{conv}	Disposition
Early strain probe	$2 \times \text{C}_{60}$	5	Superseded
Single-molecule doping	C_{60}	11	Superseded
Early electronics	$2 \times \text{C}_{60}$	24	Superseded
Exploratory design grid	$4 \times \text{C}_{60}$	20	Archive only

TABLE V. Ionic-relaxation benchmark for the tetramer P -substitution four-corner grid (fixed-cell geometry optimization: pristine and P at $\epsilon=0$ and $+3\%$, same biaxial cell). Relaxed corners and the matched-functional ratio use PBE+D3 ($\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}} = +1.19$ meV/atom; $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}| \approx 1.9$; sign not preserved). $\mathcal{S}$ follows Eq. (2) with $N=240$ atoms.

Quantity	Rigid (PBE+D3 matched)	Relaxed (geometry opt.)
$E(\text{pristine}, \epsilon=0) (E_{\text{h}})$	—	−1363.9551
$E(\text{pristine}, \epsilon=3\%) (E_{\text{h}})$	—	−1363.9550
$E(P, \epsilon=0) (E_{\text{h}})$	—	−1374.4415
$E(P, \epsilon=3\%) (E_{\text{h}})$	—	−1374.4615
$\mathcal{S}$ (meV/atom)	+1.19	−2.28
Sign of $\mathcal{S}$ vs. rigid	+	− (not preserved)

tion and may include enhanced periodic-image coupling at the smallest supercell, not a separate materials constant. Internal protocol audits bound $\sigma_{\mathcal{S}} \lesssim 2$ meV/atom on the $n \leq 4$ grid (Sec. IIC).

a. Operational significance thresholds (this work). We flag additive failure when two *adopted* criteria are met on the audited grid: (i) $|\mathcal{S}| > \sim 2$ meV/atom, above the conservative four-corner protocol band in Table I, and (ii) $\eta = |\mathcal{S}|/|\Delta E_{\text{coupled}}| > \sim 1\%$ on the same supercell. These cutoffs are operational heuristics for this PBE+D3 Γ -only production map, not universal thermodynamic boundaries or 95% confidence classifications; formal significance bands that include pending k -point and meta-GGA rows in Table I remain future work. At $n=4$ (Table II), $\mathcal{S} = -6.2 \pm 2$ meV/atom (B), $+3.0 \pm 2$ meV/atom (N), and -23.7 ± 2 meV/atom (P), where $\pm$ denotes a conservative protocol upper bound. Within $n \leq 4$, $|\mathcal{S}|$ is not monotonic in n ; we report protocol values without extrapolating $\mathcal{S}_{\infty}$. Rows with $n \geq 6$ (350 Ry) appear in Supplemental Material Table S1. Negative $\mathcal{S}$ (B, P at $n=4$) means the coupled corner lies below the additive estimate under the stated boundary conditions; positive $\mathcal{S}$ (N) means the coupled corner lies above it. Panel (e) shows one illustrative rigid-vs.-relaxed pair for $n=1$ P at $+3\%$ ($\mathcal{S}_{\text{rig}} \approx +31.9$ meV/atom vs. near-zero $\mathcal{S}$ after ionic relaxation; Table V); the near disappearance of $|\mathcal{S}|$ upon relaxation is *consistent with* a dominant elastic-work contribution under constrained load, not a direct energetic decomposition (Mayer/Bader or virial partitioning remains pending; Sec. IVF). At $n=1$ and $+3\%$, $|\mathcal{S}|$ increases monotonically with $|\Delta r_{\text{cov}}|$ for B and P (7 and 30 pm $\rightarrow$ 8.4 and 31.9 meV/atom) but N (6 pm) breaks a simple radius-only scaling (2.3 meV/atom), consistent with a secondary electronic channel (Fig. 4(d); Sec. IV B). Table V lists the matched-functional tetramer benchmark

($\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}} = +1.19$ vs. $\mathcal{S}_{\text{relaxed}} = -2.28$ meV/atom; sign not preserved).

B. Electronic structure under strain and doping

Figure 2(a) shows pristine and B/N/P-substituted tetramer cages at $\epsilon=0$ with highlighted substituent sites; covalent-radius excess Δr_{cov} is listed in Table VII. Panels (b)–(d) report one-parameter slices used in sequential screens: the N π -DOS waterfall shifts with biaxial strain; $E_g(\epsilon)$ paths place B ($E_g \approx 0.03$ eV), N ($E_g \approx -0.14$ eV), and P ($E_g \approx 0.05$ eV) at $\epsilon=0$ on distinct trajectories while pristine remains open; panel (d) gives the pristine mechanical $\Delta E(\epsilon)$ path on the reference strain grid. Supplemental Material gives full π -DOS stacks and periodic VBM/CBM isosurfaces (N size series; P $n=4$): frontier density remains cage-localized for $n = 2-6$.

C. Mechanical strain sensitivity

Pristine tetramers yield $|\alpha_{\text{pris}}| \approx 1$ meV/% on the six-point reference grid. Least-squares slopes for B, N, and P are $+58.7$, -297.5 , and $+456.0$ meV/% [Fig. 3(d); horizontal bars give linear-fit standard errors on that grid].

Table VII reports $\bar{d}$, $\sigma(\bar{d})$, and Δr_{cov} at representative strains. At $+3\%$, P shows $\Delta \bar{d} \approx 3 \times 10^{-5}$ Å versus $\sim 8.5 \times 10^{-3}$ Å for B; on the response plane in Fig. 4, P carries the largest $|\mathcal{S}|$ bubble and is the only dopant with $|\Delta r_{\text{cov}}| \gtrsim 20$ pm.

Figure 3 traces microscopic strain paths on the reference map. Hirshfeld $q(\epsilon)$ and $\bar{d}(\epsilon)$ [panels (a,b)] vary with dopant and strain. Local disorder $\sigma(\bar{d})(\epsilon)$ [panel (c)] broadens for B and N under tension while remaining near the $\epsilon=0$ baseline for P. Panel (d) maps reference-map $|\alpha|$

TABLE VI. Archived alternate-placement tetramer grid (PBE+D3 rigid; seed 137). Not used for main-text figures or quantitative claims; retained for reproducibility of the placement-sensitivity audit. Reference-map α and $S_{+3\%}^{\text{tet}}$ appear in Fig. 3(a,b). Periodic $S(n)$ is in Table II.

Dopant	Placement	α (meV/%)	$S_{+3\%}^{\text{tet}}$ (meV/atom)
B	alternate	+21.4	-2.6
N	alternate	+45.4	+12.6
P	alternate	+989.6	+41.4

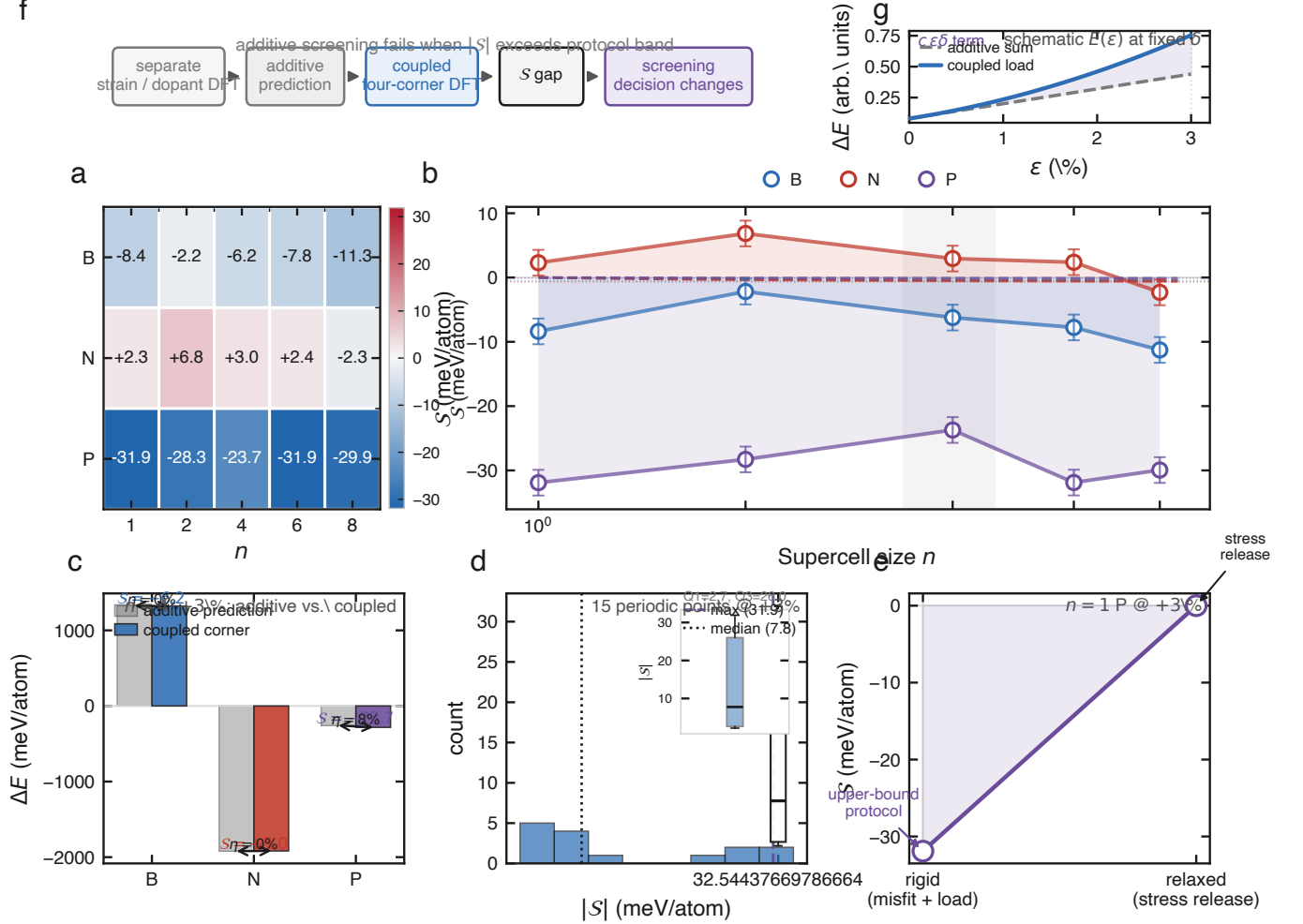


FIG. 1. When additive strain-dopant screening fails (graphical summary and audit). (f) Workflow: separate strain-only and substitution-only DFT increments are summed additively, compared with the explicitly coupled four-corner total; the gap S [Eq. (3)] flags when stability rankings change at meV-per-atom margins. (g) Schematic total-energy path $E(\epsilon)$ at fixed composition: the bilinear $c\epsilon\delta$ term bends the coupled load away from the additive sum (not from DFT data). (c) At $n=4$ and $+3\%$: additive prediction vs. coupled four-corner ΔE ; arrows label S and fractional error η [Eq. (6)]. (a,b) $S(n, \delta)$ heatmap and size-scaling trajectories with provisional $S_\infty + A/n$ guides; vertical bars in (b) show the conservative ± 2 meV/atom reporting band [Eq. (5); Table I]. (d) Distribution of $|S|$ over fifteen periodic $+3\%$ points (histogram; box: median and quartiles; max 31.9 meV/atom marked). (e) Illustrative rigid vs. relaxed path for $n=1$ P only ($+3\%$): stored elastic work under constrained load vs. stress release after ionic relaxation.

against periodic $|S|_{n=4}$ (marker area $\propto |S|_{n=1}$): N lies in the high- $|\alpha|$ /low- $|S|_{n=4}$ quadrant and P in the high- $|S|$ region—a qualitative rank inversion among three dopants ($n=3$), not a fitted correlation law.

Among audited substituents, P has the largest

covalent-radius mismatch ($\Delta r_{\text{cov}} = +30$ pm; Table VII) and the largest audited $|S|$ under the rigid loading protocol; Sec. IV B decomposes that pattern.

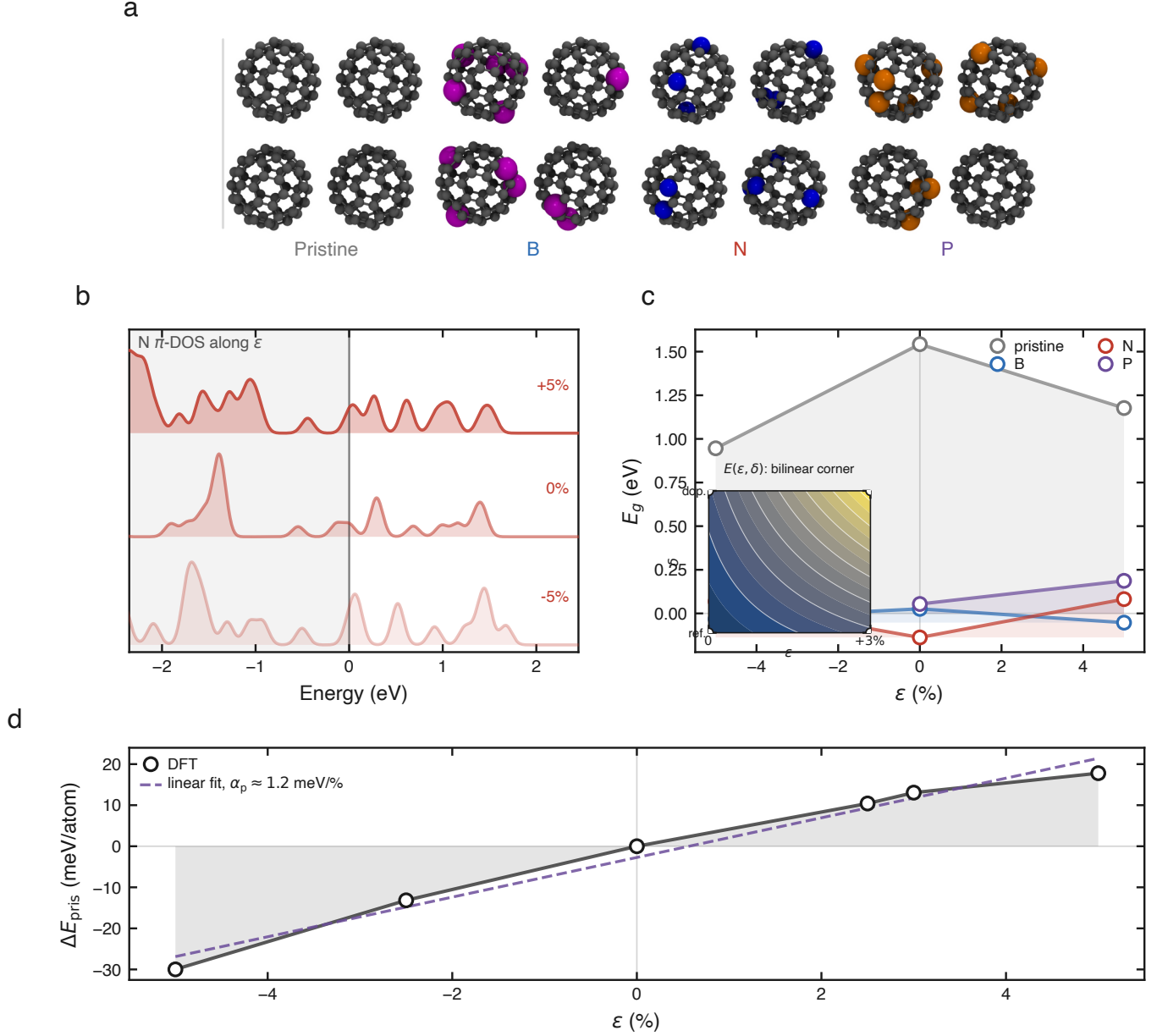


FIG. 2. Reference tetramer structure and one-parameter strain slices (PBE+D3). (a) qHP C₆₀ cages at $\epsilon=0$: pristine and B/N/P substitution sites (Δr_{cov} in Table VII). (b)–(d) Sequential-screen baselines on the reference map: N π -DOS waterfall; $E_g(\epsilon)$ with schematic $E(\epsilon, \delta)$ bilinear corner (inset); pristine mechanical $\Delta E(\epsilon)$. Joint-load four-corner audits of the omitted cross term appear in Fig. 1.

IV. DISCUSSION

A. Main finding

The present results indicate that additive strain-dopant screening fails under mechanically constrained joint load because frozen elastic incompatibility introduces a bilinear strain-composition interaction comparable to meV-per-atom stability margins on the $E(\epsilon, \delta)$ landscape [Eq. (1)]. For phosphorus at +3% biaxial load, that interaction is large enough to change high-

throughput decisions ($|\mathcal{S}| = 31.9 \pm 2$ meV/atom at $n=1$; 23.7 meV/atom at $n=4$; Table II) while linear α does not rank $|\mathcal{S}|$ or its sign [Fig. 3(d)]. $\mathcal{S}$ estimates $\partial^2 E / \partial \epsilon \partial \delta$ on the audited grid—a finite-difference diagnostic of additive failure, not the continuum coefficient γ itself (Sec. II C; Supplemental Material, Sec. I). Ionic relaxation nearly extinguishes $|\mathcal{S}|$ (Table V; Fig. 1(e)), consistent with a dominant elastic-work contribution under constrained load rather than a claim of exclusively electronic or exclusively elastic origins without partition analysis (Sec. IV F). Operationally, $|\mathcal{S}| > \sim 2$ meV/atom

TABLE VII. Local dopant–cage structure from archived *tetramer* XYZ ($\sim 5\%$ nominal doping; qualitative proxy for mismatch trends). Mean nearest-neighbor C distance $\bar{d}$ in Å; standard deviation $\sigma(\bar{d})$ across dopant sites; covalent-radius excess vs. C in pm. Strain-induced change $\Delta\bar{d} = \bar{d}(\epsilon = +3\%) - \bar{d}(\epsilon = 0)$ tracks the uniform nearest-neighbor stretch component. Distances are DFT-derived from archived tetramer XYZ; no experimental bond errors are quoted—comparison is internal across dopants and with qHP literature on intrinsic cage strain [10, 13].

Dopant	$\bar{d}(\epsilon=0)$ (Å)	$\bar{d}(\epsilon = +3\%)$ (Å)	$\Delta\bar{d}$ (Å)	$\sigma(\bar{d})$ @ +3% (Å)	Δr_{cov} (pm)
B	1.384	1.393	+0.0085	0.019	+7
N	1.393	1.402	+0.0087	0.024	−6
P	1.384	1.385	+0.00003	0.0004	+30

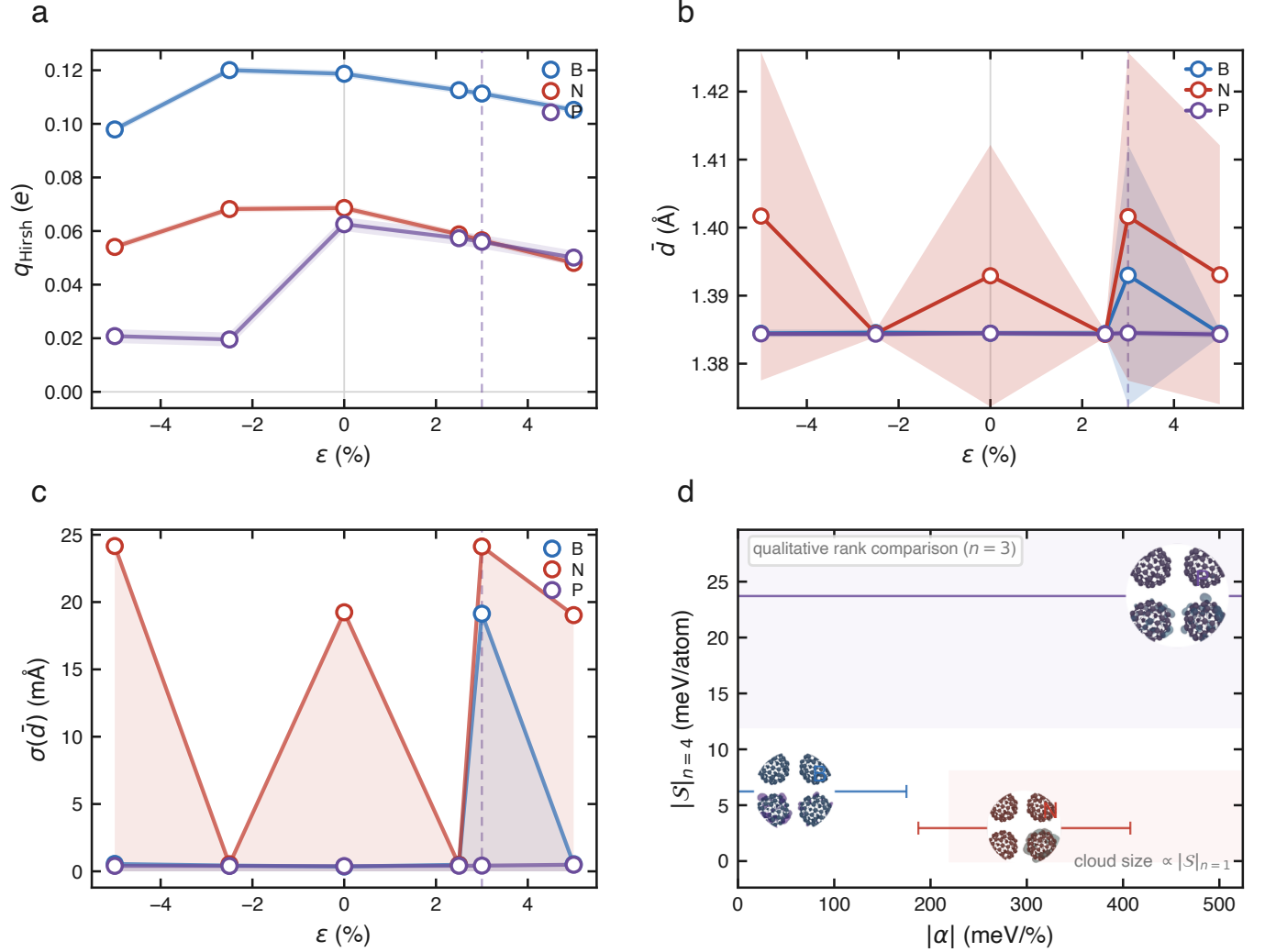


FIG. 3. Microscopic strain paths and α - $|S|$ rank comparison ($n=3$ dopants). (a) Hirshfeld $q(\epsilon)$ for B/N/P (six-point paths; dotted guide at +3%). (b) Nearest-neighbor $\bar{d}(\epsilon)$ with $\pm\sigma$ bands. (c) Local disorder path $\sigma(\bar{d})(\epsilon)$. (d) Reference-map $|\alpha|$ vs. periodic $|S|_{n=4}$ (marker area $\propto |S|_{n=1}$; horizontal bars: linear-fit standard errors on the six-point strain grid). N inverts a naive $|\alpha| \rightarrow |S|$ expectation; with only three dopants this panel reports qualitative ranking, not Pearson, R^2 , or Spearman statistics.

and $\eta > \sim 1\%$ (Sec. III A) flag cases where coupled re-evaluation is advisable on the present protocol map; these are practical screening heuristics, not thermodynamic boundaries.

B. Mechanism

For size-mismatched P, the dominant chain is (Fig. 4): $\Delta r_{\text{cov}} = +30$ pm $\rightarrow$ frozen misfit stress $\rightarrow$ overlap with biaxial tension $\rightarrow$ bilinear elastic work $\rightarrow$ largest $|S|$. *Primary* control is geometric misfit; *secondary* is elec-

tronic redistribution (Hirshfeld Δq on P $\sim 11\%$ fractional $|\Delta q/q|$, below N's $\sim 18\%$); *tertiary* is π -DOS morphology. Electronic structure responds rather than controls: π -DOS does not, by itself, rank periodic $|\mathcal{S}|$ at $n=4$ [Figs. 2(b,c) and 3(d)]. On the audited triad, $|\mathcal{S}|$ rises with $|\Delta r_{\text{cov}}|$ for B and P at $n=1$ (7 and 30 pm $\rightarrow$ 8.4 and 31.9 meV/atom); N breaks a radius-only trend (6 pm, 2.3 meV/atom). B shows the largest $\Delta \bar{d}$ at $+3\%$ yet modest $|\mathcal{S}|$; P shows the opposite (Table VII). Ionic relaxation releases misfit stress cooperatively with external tension (Fig. 1(e)); Grimme D3 contributes $\mathcal{S}_{\text{vdW}} \approx +0.33$ meV/atom at tetramer P corners (Table V), far below $|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}|$.

At $n=4$ and $+3\%$, $|\mathcal{S}| = 23.7$ meV/atom for P vs. 6.2 meV/atom for B—sequential additive estimates differ by approximately a factor of four under the same protocol.

C. Relation to literature

The present work reports a periodic *total-energy* audit of joint biaxial (ϵ, δ) load on qHP C₆₀ (Table II), rather than strain-only, doping-only, mobility, or gap-centric mapping [1, 2, 4–7, 9, 10, 12, 13, 17–22]. Cluster-expansion and configurational-interaction formalisms already encode composition–strain cross terms ($c\epsilon x$ in effective Hamiltonians) [3]; the present contribution is not a new prediction that such terms exist, but a *matched four-corner total-energy audit* on a soft, pre-strained covalent molecular network at periodic scale, with explicit rigid-vs.-relaxed and placement benchmarks (Tables V–II; Supplemental Material, Table I). Prior graphullerene studies often relaxed ions before comparing totals or combined separate grids additively [5, 7, 9], masking transient nonlinear coupling under mechanically constrained conditions (Sec. IV G 0c). The audit is not claimed unique to qHP C₆₀; it is a transferable *protocol* on soft covalent networks with local substitution, finite frozen strain, and incomplete relaxation (Sec. IV D), validated here on one host only.

D. Applicability and generality

The formalism targets hosts where the mixed derivative $\partial^2 E / \partial \epsilon \partial \delta$ can be sampled on a four-corner grid. *Applicable when*: (i) local substitutional strain is present and cannot fully relax (size mismatch, rigid substrate, or short loading times); (ii) bonding is short-range and elastically soft, so defect fields overlap under external load; (iii) the host is a covalent network with intrinsic pre-strain or elastic anisotropy [13, 23]; and (iv) ionic relaxation is incomplete so stored elastic work remains. *Not applicable when*: structures are fully relaxed bulk references with negligible frozen internal stress; metallic or liquid phases where substitutional labels are ill-defined; or high-temperature regimes where configurational entropy

dominates sub-meV energy differences (Sec. IV F). Production calculations use equibiaxial $+3\%$ load; uniaxial loading is expected to preserve the four-corner formalism but rescale the coupling coefficient through elastic anisotropy—a natural extension not audited here.

E. Experimental signatures

Laboratory probes of the stress-field picture need not resolve $\mathcal{S}$ directly but can test *additive failure*: Raman or infrared cage-mode shifts that deviate from separate strain-plus-doping calibrations; nanoindentation or substrate-bending protocols that impose strain while dopant chemistry is fixed; scanning-tunneling spectroscopy of local density of states that shifts without proportional mean bond-length change; X-ray photoelectron chemical shifts inconsistent with additive composition–strain models; and transmission-electron strain mapping combined with known substitutional loading (Table II, at.% column). Quantitative links require matched chemistry, strain calibration, and awareness that rigid-lattice audits bound constrained rather than fully relaxed configurations.

F. Limitations

Main-text α and $\mathcal{S}$ use PBE+D3 under the rigid-strain protocol; ionic relaxation can reverse or nearly extinguish $\mathcal{S}$ (Table V). Eq. (1) and the four-corner link [Eq. (4)] assume small perturbations on a single smooth energy branch; they lose validity at large strain (e.g. $\gtrsim 10\%$) or across structural phase transitions, and the production audits intentionally remain at $+3\%$ biaxial load inside the nominally elastic regime. All totals are 0 K Born–Oppenheimer energies; finite-temperature free energies and configurational entropy remain unexplored and could reorder sub-meV competitors without restoring additivity if bilinear coupling persists at the electronic-structure level. Production periodic cells use Γ -only sampling; dense- k and SCAN/r²SCAN spot checks remain targeted backlogs (Table I). Alternate placements (Table II) bound configuration specificity; Table VII geometry is a tetramer proxy (~ 5 at.%) without Mayer/Bader partitioning or matched periodic $n=4$ bond statistics. With only three substitutional labels, the α – $|\mathcal{S}|$ map [Fig. 3(d)] is a qualitative rank comparison among $n=3$ chemistries, not a fitted correlation law or predictive model for linear α . Direct elastic-vs.-electronic energetic decomposition (Mayer bond order, Bader strain partitioning, virial stress) and dispersion cross-term audits for B/N periodic cells remain open (Table I).

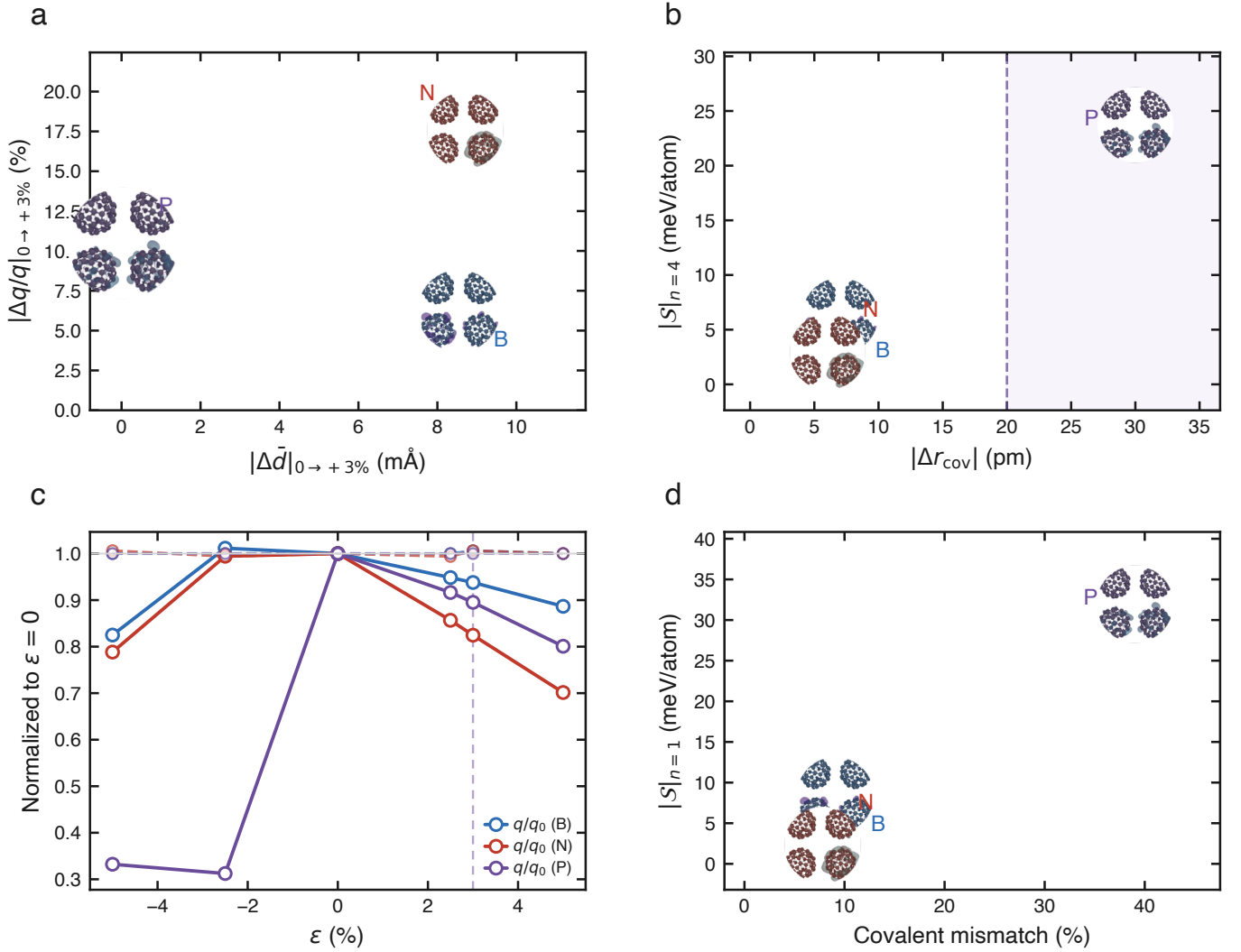


FIG. 4. Mechanism hierarchy for P-dominated coupling (primary: misfit strain $\rightarrow$ frozen internal stress; secondary: charge redistribution; tertiary: π -DOS morphology). (a) Response plane from 0 $\rightarrow$ +3% path endpoints; bubble area $\propto |S|_{n=4}$ (B: large $\Delta \bar{d}$, small $|S|$; N: large $|\Delta q|$, moderate $|S|$). (b) $|\Delta r_{cov}|$ vs. $|S|_{n=4}$: radius alone does not rank B/N/P. (c) Linked normalized paths $q(\epsilon)/q(0)$ (solid) and $\bar{d}(\epsilon)/\bar{d}(0)$ (dashed). (d) Covalent mismatch (%) vs. $|S|_{n=1}$ (P extremum at largest mismatch).

G. Outlook, workflow, and predictions

a. Citation workflow. For a candidate host: (1) relax or fix coordinates according to the loading scenario; (2) compute separate strain-only and substitution-only increments; (3) evaluate the four-corner $\mathcal{S}$ at the target (ϵ, δ) ; (4) if $|S| > \sim 2$ meV/atom or $\eta > \sim 1\%$, replace sequential screening with explicit coupled DFT on the same functional and cell (Fig. 1(f,g)). Rigid-strain $\mathcal{S}$ bounds epitaxial clamping, substrate adhesion, ultrafast loading, or nanoscale flexures where relaxation is incomplete (Sec. IV G 0 c).

b. Prediction. Within the audited covalent-radius trend, larger-radius group-V substituents such as As or Sb are *expected* to exhibit stronger bilinear coupling under constrained loading than P, because frozen misfit

stress should scale with $|\Delta r_{cov}|$; this extrapolation is not verified by present calculations. Targeted follow-up includes Mayer/Bader strain-path partitioning, $\mathcal{S}(\text{cutoff})$, $\mathcal{S}(k)$ sensitivity curves, and periodic $n=4$ P ionic relaxation on the production map.

c. Physical relevance of mechanically constrained loading. Two loading regimes must be distinguished. *Constrained (rigid-lattice) audits* bound epitaxial clamping, substrate adhesion, ultrafast loading, or nanoscale flexures where internal coordinates cannot fully relax; here $|S|$ can reach tens of meV/atom (periodic $n=1$ P at +3%: -31.9 meV/atom rigid vs. ~ 0.1 μeV /atom relaxed at matched PBE+D3; Table V). *Equilibrium (fully relaxed) workflows*—slow thermal strain on unclamped films, post-growth annealing, or standard geometry-optimized high-throughput screens—are expected to re-

cover near-additive behavior when misfit stress is released (Fig. 1(e)); additive sequential screening may remain adequate there, and the present rigid-map magnitudes must not be read as equilibrium free-energy differences. Rigid-strain $\mathcal{S}$ bounds omitted coupling when internal stress remains frozen; Fig. 1(e) contrasts stress preservation under rigid load with release upon relaxation.

V. CONCLUSIONS

Nonlinear strain–dopant energetic coupling—not the symbol $\mathcal{S}$ —is the primary finding: additive composition–strain screening fails under constrained load on qHP C₆₀ because stored elastic incompatibility generates a bilinear interaction large enough to reorder stability rankings at meV-per-atom margins ($|\mathcal{S}| > \sim 2$ meV/atom and $\eta > \sim 1\%$; Sec. III A). The matched four-corner difference $\mathcal{S}$ quantifies that failure [Eqs. (4)–(3)] as a discrete mixed-derivative estimator, not a new thermodynamic observable. On qHP C₆₀, donor/acceptor dopants produce an α sign split [Fig. 3(d)] that does not rank $|\mathcal{S}|$ among the three audited species, while periodic $|\mathcal{S}|$ reaches 31.9 ± 2 meV/atom for P at $n=1$ at fixed coordinates—an extremum within a broader 2.2–31.9 meV/atom audit distribution (Fig. 1(d)) with $\eta \approx 8\%$ at $n=4$. Non-additivity correlates with frozen local elastic incompatibility on the B/N/P triad rather than with dopant identity alone; for P, geometric misfit strain is primary, electronic redistribution secondary, and π -DOS morphology tertiary (Sec. IV B; Fig. 4). Ionic relaxation nearly extinguishes $|\mathcal{S}|$, consistent with a dominant elastic-work contribution under constrained load rather than an exclusive electronic-interaction mechanism (Table V; Fig. 1(e)).

Coupled (ϵ, δ) screening is advisable for $|\Delta r_{\text{cov}}| \gtrsim 20$ pm substituents under *constrained* +3% loads; four-corner audits can pre-screen such cases before full relaxation (Sec. IV G). Generalization beyond qHP C₆₀ requires matched four-corner audits on each host; the present study reports failure conditions and an audit protocol on one soft covalent molecular network and does not claim universal constants, second-host benchmarks, or materials-general laws. Prior strain–doping studies often masked this coupling by fully relaxing structures before comparing totals or by assembling separate strain-only and doping-only scans additively (Sec. IV C). The theoretical landscape is $E(\epsilon, \delta)$; $\mathcal{S}$ is its quantitative diagnostic under stated boundary conditions, not the primary physical claim.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

The data that support the findings of this study are openly available at <https://github.com/chenxingqiang/graphullerene-strain-engineering>, including computational inputs and outputs, tabulated coupling-energy and polaron analyses, four-corner total-energy components underlying $\mathcal{S}$ (machine-readable synergy tables), and materials to reproduce all figures and tables in the main text. Ionic-relaxation, Hirshfeld, and size-scaling archives referenced in Tables VI–II are included in the public release. A tagged release matching the manuscript revision will be issued upon acceptance.

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